CS 422 Assignment 4
Part 1: Uniform Traffic Matrix

Kush, Joseph, Issac

Main idea

Every node sends $\frac{d}{n-1}$ to each of the other $n-1$ nodes, so each node sends d in total and the whole network has to carry nd. The network also has exactly nd capacity (n nodes, d links each, capacity 1). So the demand fits only if every flow takes 1 hop. A flow that takes 2 hops uses 2 units of capacity, a flow that takes 3 hops uses 3, and so on. This gives

$$\theta \cdot \frac{d}{n-1} \sum_{s \neq t} \text{dist}(s,t) \leq nd \quad \Longrightarrow \quad \theta \leq \frac{1}{\text{average shortest path length}}$$

So to maximize concurrent flow, I need a topology where the average number of hops is as small as possible.

How short can paths be?

From any node, with d outgoing links I can reach at most d nodes in 1 hop, at most d^2 in 2 hops, d^3 in 3 hops, and so on. So the best possible graph spreads out like a tree, where every hop multiplies the number of nodes I can reach by d and no paths overlap. This gives a lower bound on the average path length, and so an upper bound on θ.

My topology

To get this tree-like spreading, I think of each node as a number in base d. Each node connects to the nodes I get by shifting its digits left and adding a new last digit. Every hop adds one digit, so I can reach any node in about $\log_d n$ hops. Written with mod n so it works for any n:

1. For every node $i = 0, \ldots, n-1$ and every $j = 1, \ldots, d$, add a link $i \to (-d \cdot i - j) \bmod n$. (The minus sign just relabels the nodes; it gives fewer self-loops.)
2. If some node a has a link to itself, pick another link $u \to v$ (not touching a, and where $u \to a$ and $a \to v$ do not already exist) and replace $a \to a$ and $u \to v$ with $u \to a$ and $a \to v$. Every node still has d links in and d links out.

Why this is the best

- θ is limited only by how many hops the flows take, so the best topology is the one with the shortest average path.

- My graph grows by about a factor of d every hop, so every node is about $\log_d n$ hops from every other node. That is the same as the tree bound above, the best any graph can do.
- A perfect tree, where no paths ever overlap, is impossible for $d \geq 2$ (directed Moore graphs do not exist). So no topology reaches the bound exactly, and the best possible topology is one that gets as close as it can. I checked my graph for all $n < 120$ with $d \mid n - 1$, and its average path length was always within about 14% of the bound.

Example: $n = 9$, $d = 2$

Node i links to $(-2i - 1) \bmod 9$ and $(-2i - 2) \bmod 9$:

node	links to
0	8, 7
1	6, 5
2	4, 3
3	2, 1
4	0, 8
5	7, 6
6	5, 4
7	3, 2
8	1, 0

Average path length is 2.17, and the best possible is 2.0, so $\theta \leq 1/2.17 \approx 0.46$, close to the best possible 0.5.